

Blood Smear Examination

Giemsa's Staining Method

- 1) Fix the blood smear in methanol for 2-3 minutes
- 2) Prepare 1 in 10 dilution of *Giemsa's* stain in Phosphate Buffer (pH-6.8) or Distilled Water
- 3) Cover the smear with diluted *Giemsa's* stain for 15-20 minutes
- 4) Wash the smear with distilled water and air dry
- 5) Examine under oil immersion objective

Leishman's Staining Method

a) Procedure-

- 1) Cover the unfixed blood smear with *Leishman's* stain (5-10 drops) for 1-2 minute
- 2) Add double quantity of phosphate buffer or distilled water and mixed by blowing air
- 3) Let the diluted stain act for 10-15 min
- 4) Wash the slide with distilled water and air dry
- 5) Examine under oil immersion objective

b) Examination of slide-

- 1) Set the smear under microscope under low power
- 2) Put the drop of cedar wood oil and set the smear under oil immersion lens
- 3) Count at least 100 leucocytes by battlement method

c) Interpretations-

- 1) *Neutrophillia* is seen in localized bacterial infections, stress, blood loss and haemolytic conditions
- 2) *Eosinophillia* indicates allergic condition or parasitic infestation
- 3) *Monocytosis* is encountered in chronic infection
- 4) *Lymphocytosis* represents chronic infection